AJKD
Atlas of Renal Pathology

Amyloid

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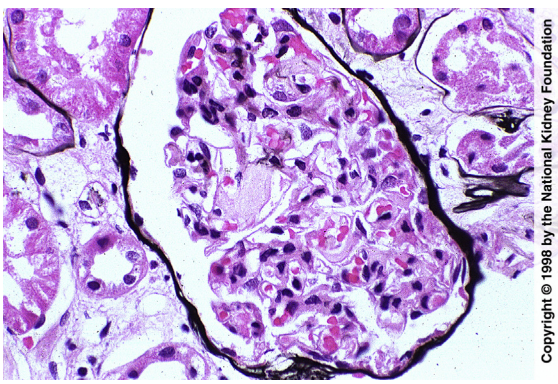

Fig 1. Acellular eosinophilic material is typical of amyloid deposits and may be found in mesangial areas or peripheral capillary loops within the glomerulus, as in this case (Jones silver stain; original magnification ×200).

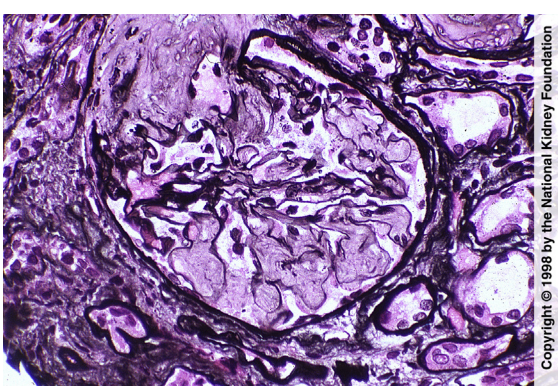

Fig 2. In a severe case of amyloidosis, massive distention of mesangial areas and distortion of peripheral capillary loops are secondary to amorphous, acellular, slightly eosinophilic pale material, which in this case also involves the arteriole at the vascular pole. This material has a more fluffy appearance than the sclerosis or mesangial matrix expansion that can occur in other conditions (Jones silver stain; original magnification ×200).

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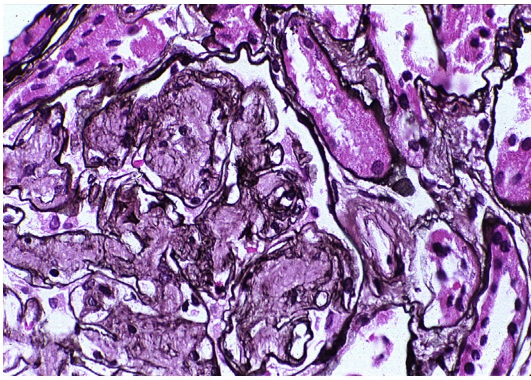
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Fig 3. On high-power view of amyloid, basement membrane irregularities and feathery spikes are evident, in addition to amyloid infiltration in mesangial areas and in the arteriole to the right (Jones silver stain; original magnification ×400).

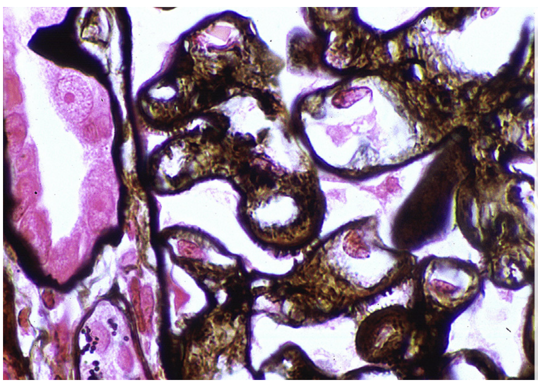
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Fig 4. On high power, irregular feathery basement membrane spicules may be seen in cases of amyloid, due to the infiltration of amyloid irregularly through the basement membrane, as in this case (Jones silver stain; original magnification ×1,000).

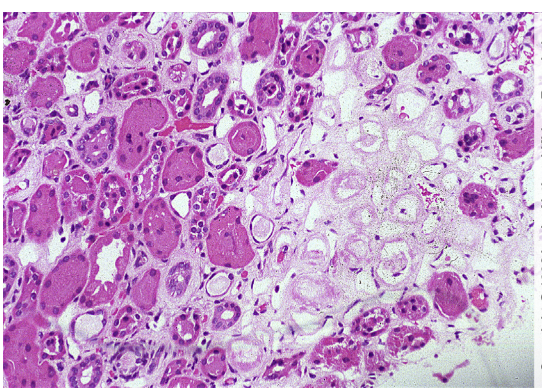
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Fig 5. Amyloid may also involve the tubular interstitium, resulting in acellular pink areas that could be mistaken for areas of necrosis, as illustrated here (hematoxylin & eosin; original magnification ×100).

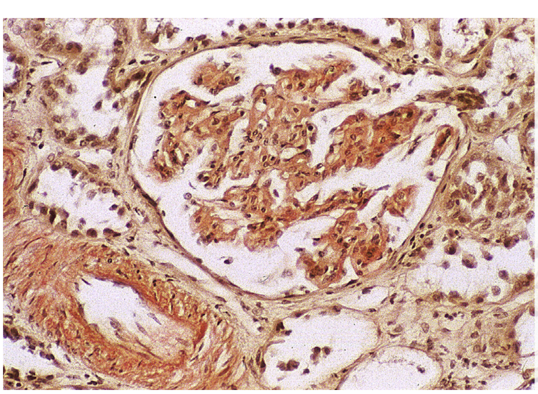
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Fig 6. The gold standard for diagnosis of amyloid is Congo red positivity, shown here, which also must show apple green birefringence under polarized light (see next slide). The Congo red positivity is present in the mesangium, in the small artery, and along the basement membrane. (Congo red stain; original magnification ×200).

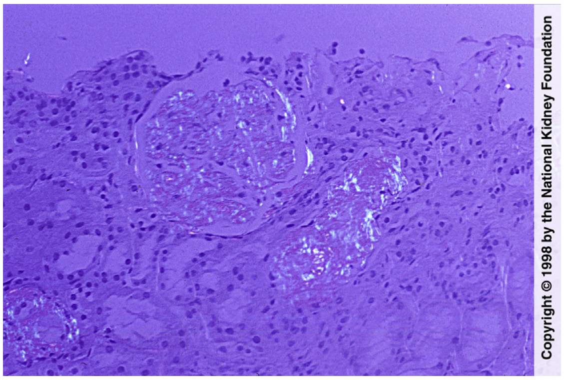
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Fig 7. When the Congo red stain is viewed under polarized light, areas of amyloid show apple green birefringence. In this case, amyloid deposits are seen in the mesangial areas, the capillary loops, and in vessels (Congo red stain; original magnification ×100).

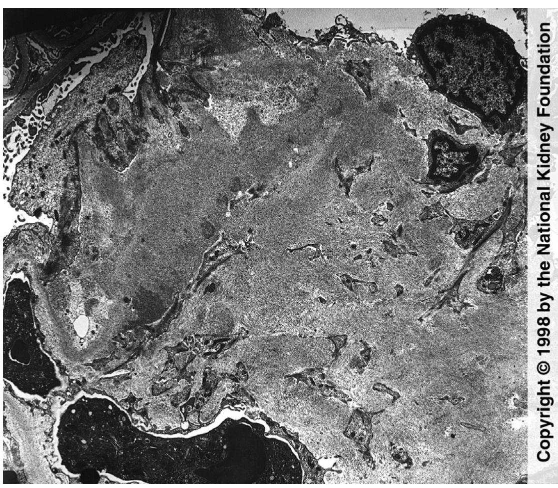
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Fig 8. Electron microscopy shows massive expansion of the mesangium by fibrillar material with randomly oriented thin fibrils with a diameter of 10 to 12 nm, often extending to basement membranes as in this case (transmission electron microscopy; original magnification ×8,000).

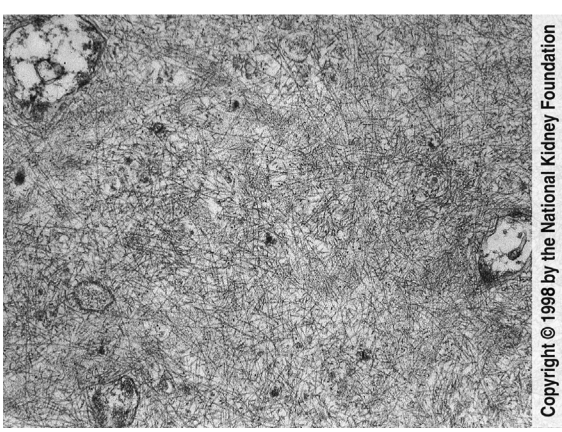
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Fig 9. By electron microscopy, amyloid appears as randomly oriented thin fibrils, 10 to 12 nm in diameter, with a loose, flocculent background (transmission electron microscopy; original magnification ×51,250).

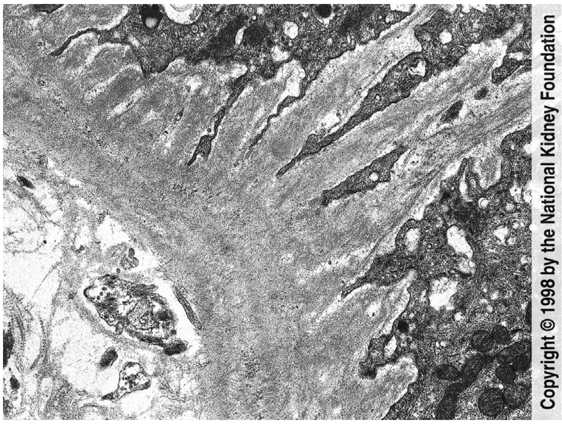

Fig 10. Amyloid infiltration through the basement membrane with resulting feathery spikes with basement membrane material and delicate amyloid fibrils are shown in this case (transmission electron microscopy; original magnification ×20,250).